

Lab 4: WCAG 2 AA: Understanding the Web Content Accessibility Guidelines (WCAG)

Introduction: Making digital content usable for everyone, including those with impairments, is a crucial component of web design. For locating and fixing accessibility flaws, the Web Content Accessibility Guidelines (WCAG) offer a strong foundation. We can enhance digital platforms' inclusivity and usability by examining accessibility problems in the real world and connecting them to the pertinent WCAG success criteria.

Objectives of this lab:

- Determine and evaluate digital content's accessibility flaws.
- Connect these flaws to the associated WCAG failure modes, methods, and success criteria.
- Inform people about accessibility concerns based on WCAG guidelines.
- Describe how these flaws affect users and offer fixes based on the best accessible standards.

	Accessibility defect	Failure of Success Criteria	Sufficient technique (General, HTML) or failure	Impacted users
EX	(Example) The HR diagram does not have any text alternative	1.1.1	H37: Using alt attributes on img elements	Screen reader users will not be aware of the information on the graphics
1	The heading "Your leave balance" has been styled to look like a heading but has not been marked up as a heading within the HTML code.	1.3.1 (Info and Relationships)	G141: Organizing a page using headings	Screen reader users who depend on proper heading structure for navigation.
2	The progress bar: When it receives focus, the screen reader does not announce any role (i.e., slider)	4.1.2 (Name, Role, Value)	ARIA1: Utilizing ARIA landmarks to define specific sections of a webpage	Screen reader users who need to understand the role of interactive elements.
3	The "username" label is not connected to the corresponding input field	1.3.1 (Info and Relationships), 3.3.2 (Labels or Instructions)	H44: Using label elements to associate text labels with form controls	Screen reader users who need labels to understand specific fields in the form.

	Accessibility defect	Failure of Success Criteria	Sufficient technique (General, HTML) or failure	Impacted users
4	Common input fields below lack a way to communicate purpose. -Your full name -your mobile phone -Your email address	1.3.5 (Identify Input Purpose)	G83: Providing text descriptions to identify required fields that were not identified in the user interface.	This lack of communication can make it difficult for people with disabilities to understand what information is required in these fields.
5	Heading level 3 appears after heading level 1, skipping heading level 2.	1.3.1 (Info and Relationships)	G141: Organizing a page using headings	Skipping levels can create confusion for users relying on assistive tools. Screen reader users who rely on proper heading structure for navigation.
6	“Add employee” modal dialog: When the dialog is closed, the focus goes to the top of the page instead of the trigger element on the page	2.4.3 (Focus Order)	G107: Using ‘focus’ to return the user to the trigger element	Users with cognitive disabilities may struggle with the sudden change in focus and they might lose track of where they were in the interface.
7	Instructions necessary for form completion, especially regarding required fields, are missing.	3.3.2 (Labels or Instructions)	G83: Providing text descriptions to identify required fields in the form.	Users with cognitive or visual impairments, and those using screen readers, may struggle to complete the form without clear instructions.

	Accessibility defect	Failure of Success Criteria	Sufficient technique (General, HTML) or failure	Impacted users
8	On a mobile view (320 CSS pixels), the carousel back and next buttons disappear	2.1.1 (Keyboard)	G202: Ensuring keyboard control for all functionality	The lack of visible buttons makes it difficult for users with motor impairments. And Individuals with low vision may face several challenges, including difficulty tracking pointers, which can hinder their ability to use software effectively.
9	The first paragraph is clipped with 200% zoom	1.4.4 (Resize Text)	G178: Providing controls on the Web page that allow users to incrementally change the size of all text on the page up to 200 percent	Users with age-related vision loss and some people with cognitive disabilities who require increased font size may face difficulties.
10	The color contrast ration between the text & the background in the error message is 3.1:1	1.4.3 (Contrast (Minimum))	G18: Ensuring that a contrast ratio of at least 4.5:1 exists between text (and images of text) and background behind the text	Some individuals with cognitive disabilities need low-contrast color combinations might face visual processing issues.
11	The video “Tutorial for new users” has no closed caption	1.2.2 (Captions (Prerecorded))	G87: Providing closed captions using any readily available media format that has a video player that supports closed captioning	People who are deaf or have a hearing loss will have hard time understanding the tutorial message.

	Accessibility defect	Failure of Success Criteria	Sufficient technique (General, HTML) or failure	Impacted users
12	Added to employees list: This is a status update that dynamically changes without page reload but it is missed from the alert live region content and therefore not indicated to screen reader users.	4.1.3 (Status Messages)	G199: Providing success feedback when data is submitted successfully	Screen reader users and visual impairments individuals needing notifications of changes in content.

References:

WebAIM: WebAIM's WCAG 2 Checklist. (n.d.). Webaim.org.

<https://webaim.org/standards/wcag/checklist>

How to Meet WCAG (Quickref Reference). (n.d.). Wwww.w3.org.

<https://www.w3.org/WAI/WCAG22/quickref/?versions=2.1>